

Microbial Diseases of the Respiratory System

Respiratory System Module

Bacterial causes of the Respiratory Infections

Viral and Fungal causes of the Respiratory Infections

Microbial Diseases of the Respiratory System

- **Bacterial Diseases of the Upper Respiratory System, Sinuses, and Ears**
 - Pharyngitis*
 - Diphtheria
 - Epiglottitis
 - Sinusitis and Otitis Media
- **Viral Diseases of the Upper Respiratory System**
 - Common Cold
 - Croup
- **Bacterial Diseases of the Lower Respiratory System**
 - Pneumonia*
 - Pertussis
 - Legionnaires' Disease
 - Inhalational Anthrax
 - **Tuberculosis**

Microbial Diseases of the Respiratory System...

- **Viral Diseases of the Lower Respiratory System**
 - Influenza
 - Acute Bronchitis
 - Bronchiolitis
 - Hantavirus Pulmonary Syndrome (HPS)
 - **SARS**
 - **COVID-19**
- **Mycoses of the Lower Respiratory System**
 - Coccidioidomycosis
 - Blastomycosis
 - Histoplasmosis
 - Pneumocystis Pneumonia (PCP)

Structures of the Respiratory System

- Anatomically divided into:
 - **Upper Respiratory Tract (URT)**
 - **Lower Respiratory Tract (LRT)**
- **Upper Respiratory Tract (URT): Main Structures:**
 - Nose and nasal cavity
 - **Pharynx (throat)** – nasopharynx, oropharynx, laryngopharynx
 - Paranasal sinuses
 - Epiglottis
 - **Larynx (voice box)** – often considered a transitional structure
- **Lower Respiratory Tract (LRT): Main Structures:**
 - Trachea (windpipe)
 - Bronchi
 - Bronchioles
 - Alveoli (air sacs)

Microbial Diseases of the Upper Respiratory System, Sinuses, and Ears

Pharyngitis

- Pharyngitis is inflammation of the pharynx (**throat**).
- Commonly known as a **sore throat**.
- Can be **viral** or **bacterial** in origin.
 - **Most cases** of pharyngitis are caused by **viruses**.
 - Approximately **90% of sore throats in adults** and **60–75% of sore throats in children** are caused by viruses
 - **S. pyogenes** is **the most common bacterial** cause of acute pharyngitis.
- **Causes:**
 - Viral: **Rhinovirus, Influenza, Adenovirus, Epstein-Barr virus (EBV)**
 - Bacterial: ***Streptococcus pyogenes*** (Group A Strep).
 - Allergies and irritants (smoke, pollution) may also contribute.

Pharyngitis...

Symptoms

- Sore throat
- Painful swallowing
- Red, swollen pharynx
- Fever and headache
- Swollen lymph nodes (in bacterial cases)

Pharyngitis...

- **Viral Pharyngitis:**
 - Caused by viruses such as the **Rhinovirus, influenza virus**, and others.
 - Common Symptoms: **Cough, runny nose**
 - Fever, Sore throat, Fatigue
- **Bacterial Pharyngitis**
 - Often caused by *Streptococcus pyogenes* (strep throat).
 - Common Symptoms:
 - **Severe pain upon swallowing** (generally of sudden onset).
 - **White spots** on tonsils or throat (**exudate**)
 - **Petechiae** on the roof of the mouth
 - **Gray furry tongue, Swollen uvula**
 - High fever, **swollen lymph nodes, absence of cough**

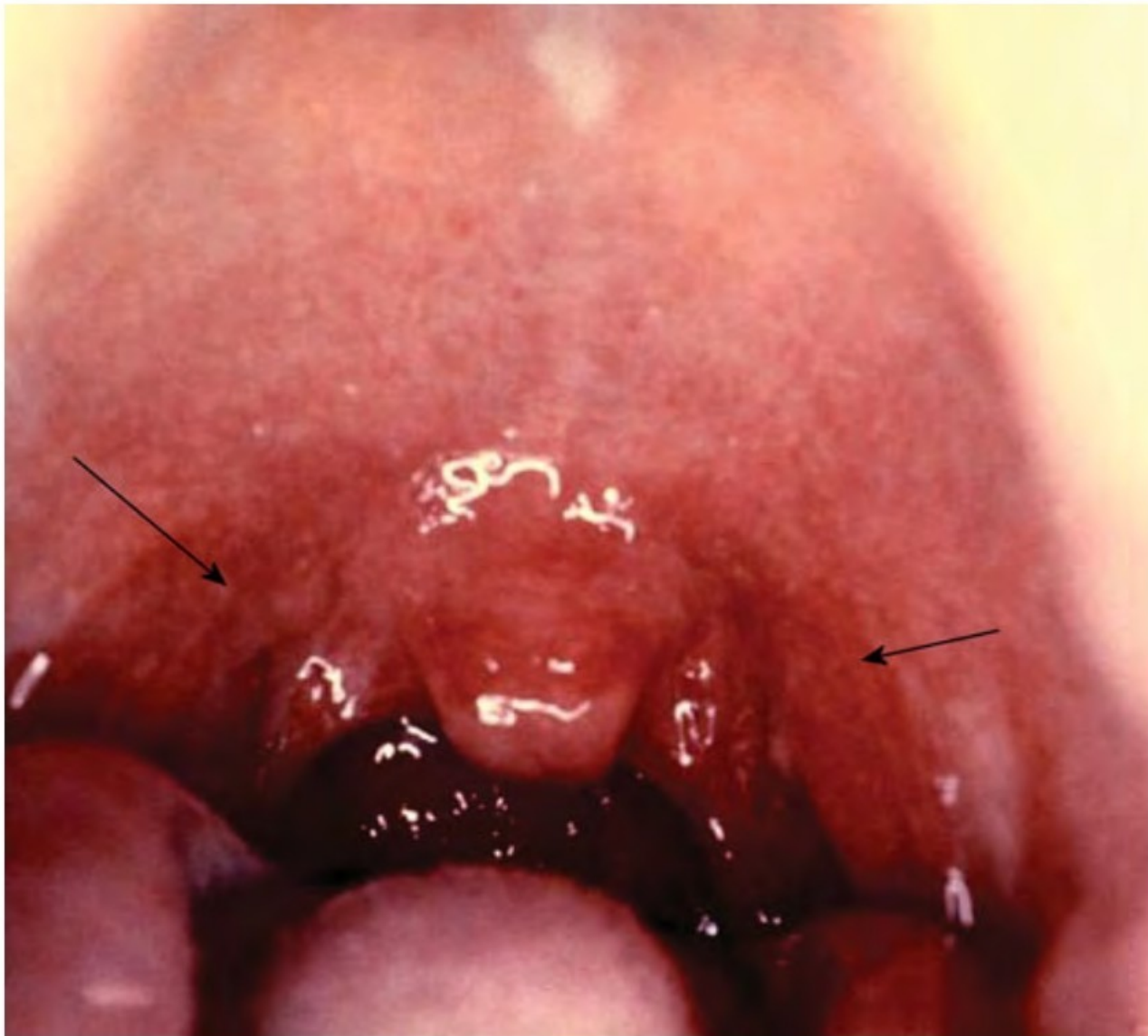


Figure. Inflammation of the **oropharynx (soft palate) and tonsils** in a patient with ***S. pyogenes* pharyngitis**. Notice the **redness** on the oropharynx and tonsils. (Note: **Viral pharyngitis** can appear similar to this image.)



Figure. The **soft palate of a patient with pharyngitis**. Notice the **petechia** (arrows) on the soft palate.

- Petechiae are tiny, **pinpoint red**, purple, or brown **spots** on the skin that result from small blood vessel bleeding.

Pharyngitis...

- **Diagnosis**
 - Clinical examination
 - **Rapid Strep Test** or **throat culture**
- **Treatment**
 - Viral: **Supportive care** (rest, fluids, pain relievers)
 - Bacterial: **Antibiotics** (e.g., penicillin or amoxicillin)
 - Symptom relief with gargles, lozenges, antipyretics
- **Complications (if untreated)**
 - Rheumatic fever
 - Post-streptococcal glomerulonephritis
 - Peritonsillar abscess



Pharyngitis...

- **Prevention**
 - Hand hygiene
 - **Avoid close contact** with infected individuals
 - Treat strep throat promptly
 - Avoid smoking and irritants
 - Prevention and early care reduce complications.

Diphtheria

- Diphtheria is an acute bacterial disease caused by *Corynebacterium diphtheriae*.
- Affects the **mucous membranes** of the **throat** and **nose**.
- Produces a **potent exotoxin** that can damage multiple organs.
- **Types of Diphtheria**
 - **Respiratory Diphtheria: Most common**, affects throat and tonsils.
 - **Cutaneous Diphtheria:** Skin infection with ulcers.
 - **Systemic complications** due to toxin absorption.
- **Transmission**
 - Spread via **respiratory droplets, close contact**, or contaminated objects.
 - Incubation period: **2–5 days**
 - Highly contagious in unvaccinated populations.

Diphtheria...

- **Clinical Features**
 - Sore throat and fever
 - **Grayish-white pseudomembrane** on tonsils and throat
 - **Difficulty swallowing**
 - Swollen neck (“**bull neck**” appearance)
 - Can lead to breathing problems and **airway obstruction**
- **Diagnosis**
 - Clinical presentation
 - **Throat swab** for **culture** and **toxin testing**
 - PCR in specialized labs



Figure. A patient with **diphtheria**. Severe **cervical lymphadenopathy and edema** cause the “**bull neck**” appearance characteristic of this disease.

Diphtheria...

- **Treatment**
 - **Diphtheria antitoxin (DAT)** to neutralize toxin
 - **Antibiotics:** erythromycin or penicillin
 - Airway support if obstruction occurs
 - Isolation to prevent spread
- **Prevention**
 - Vaccination (**DPT**)
 - Early case detection and contact tracing

Epiglottitis

- Acute inflammation of the **epiglottis** and **adjacent structures**.
- Can lead to **life-threatening airway obstruction**.
 - A medical emergency in the **upper airway**
- **Most common in children** but can occur in adults.
- **Causes**
 - Bacterial: **Haemophilus influenzae type b (Hib)** — **most common**
 - **Others**: Streptococcus pneumoniae, Staphylococcus aureus
 - Rarely viral or fungal

Epiglottitis...

- **Clinical Features**

- Sudden onset high fever
- Severe sore throat
- Difficulty swallowing (dysphagia)
- Drooling (excess **saliva** flows out of your mouth involuntarily)
- Stridor and muffled voice (“hot potato” voice)
 - **Stridor** is a **high-pitched**, whistling sound made during breathing, often indicating an upper airway **obstruction**.
 - A **muffled voice**, where the voice sounds like it's coming from **behind a mask**, can also be a sign of upper airway problems.
- Sitting in **tripod position**



Epiglottitis...

- **Diagnosis**
 - **Clinical** diagnosis is critical — do not delay treatment
 - Lateral neck **X-ray**: “**Thumb sign**” (swollen epiglottis)
 - **Avoid throat examination** in children (may trigger airway spasm)
 - Confirmed by laryngoscopy (in controlled setting)
- **Treatment**
 - Airway management is the priority
 - Intubation or tracheostomy if needed
 - **IV antibiotics**: ceftriaxone or cefotaxime
 - Supportive care: fluids, oxygen, monitoring

Epiglottitis...

- **Complications**
 - Airway obstruction
 - Respiratory arrest
 - Sepsis
 - Death if untreated
- **Prevention**
 - **Hib vaccination** — highly effective
 - Avoid delay in seeking care for sudden sore throat with breathing difficulty
 - Early intervention saves lives

Sinusitis and Otitis Media

Introduction: Sinusitis and Otitis media

- Usually occur as **complications of common cold or pharyngitis**.
- Caused by **bacteria that spread** from the **nasopharynx**.
- Common in **children and individuals with respiratory allergies**.

Sinusitis – Definition

- **Inflammation or infection of the paranasal sinuses**.
- Often follows **obstruction of sinus drainage** after viral infection.
- May be **acute** (<4 weeks) or **chronic** (>12 weeks).

Sinusitis and Otitis Media...

Sinusitis – Etiology (**Causative Agents**)

Bacterial causes:

- *Streptococcus pneumoniae*
- *Haemophilus influenzae*
- *Moraxella catarrhalis*

Other causes:

- Viral: rhinoviruses, adenoviruses
- Fungal: *Aspergillus* (in immunocompromised)

Sinusitis and Otitis Media...

Sinusitis – Clinical Features

- Nasal congestion and discharge (purulent)
- Facial pain or pressure (maxillary, frontal, etc.)
- Headache
- Fever, malaise
- Reduced sense of smell

Sinusitis – Diagnosis

- Clinical features
- Imaging (X-ray/CT) if needed
- Culture of nasal discharge (rarely required)

Sinusitis – Treatment

- Symptomatic relief (decongestants, saline irrigation)
- Antibiotics if bacterial (e.g., Amoxicillin–clavulanate)
- Surgery for chronic/recurrent cases.

Sinusitis and Otitis Media...

Otitis Media – Definition

- **Infection or inflammation of the middle ear cavity.**
- Common in **young children** due to short Eustachian tube.
- Often follows **upper respiratory viral infection**.

Otitis Media – Etiology (**Causative Agents**)

Bacterial causes:

- *Streptococcus pneumoniae* (**most common**)
- *Haemophilus influenzae*
- *Moraxella catarrhalis*

Viral causes:

- Respiratory syncytial virus (RSV), rhinovirus, influenza virus

Sinusitis and Otitis Media...

Otitis Media – Clinical Features

- Ear pain and fever
- Hearing loss or fullness in ear
- Irritability (in children)
- **Bulging tympanic membrane** on otoscopy
- Possible discharge if perforation occurs

Otitis Media – Diagnosis

- Clinical examination (otoscopy)
- Tympanometry or culture (if discharge)

Otitis Media – Treatment:

- Analgesics, antipyretics
- Antibiotics (Amoxicillin)
- Tympanostomy tube for recurrent cases

Complications: mastoiditis, hearing loss, meningitis (rare)

Common Cold

- The common cold is a **mild viral infection of the upper respiratory tract**.
 - A **self-limiting** viral upper respiratory infection
- Affects the **nose, throat, and sinuses**.
- **Most frequent infectious disease** worldwide.
- **Causative Agents**
 - **Rhinoviruses** – most common (Picornaviridae, +ssRNA virus, >100 types)
 - Others: **Coronaviruses, Adenoviruses, Respiratory syncytial virus (RSV), Parainfluenza virus**
 - **Transmission:** Airborne droplets, direct contact, and contaminated surfaces

Common Cold...

- **Clinical Features**

- Runny or stuffy nose
- Sneezing
- Sore throat
- Cough
- Mild headache or body aches
- Low-grade fever (rare in adults)

- **Diagnosis**

- Based on clinical symptoms
- No laboratory tests needed
- Rule out influenza, COVID-19, or bacterial infections if symptoms are severe

Common Cold...

- **Treatment**

- Supportive care only
- Rest, fluids, warm drinks
- Decongestants, throat lozenges, pain relievers (e.g., paracetamol)
- Antibiotics not effective (viral cause)

- **Prevention**

- **Frequent handwashing**
- Avoid close contact with infected people
- Maintain good hydration and immunity

Croup

- Croup is a **viral respiratory illness** that causes inflammation of the **upper airway**.
- Affects the **larynx, trachea, and bronchi**.
 - Acute **Laryngotracheobronchitis**
- Common in children aged **6 months to 5 years**.
- **Causes**
 - Viral: Most commonly **parainfluenza virus**
 - Others: **Influenza, RSV, adenovirus, and measles virus**.
 - Spread through **respiratory droplets**.

Croup...

- **Symptoms**

- Barking cough (seal-like)
- Inspiratory stridor
- Hoarseness
 - Also known as **dysphonia**, is a change in the quality of voice, making it sound rough, raspy, strained, or breathy
- Fever
- Symptoms worsen at **night**

- **Diagnosis**

- Based on **clinical** symptoms and age
- Neck X-ray may show “steeple sign” (subglottic narrowing)
- No routine labs required in mild cases

Croup...

- **Treatment**
 - **Mild cases:** Humidified air, hydration, rest
 - **Moderate to severe:**
 - **Dexamethasone** (oral or IM)
 - Nebulized epinephrine
 - Hospitalization if severe respiratory distress
- **Complications**
 - Airway obstruction
 - Hypoxia
 - Rarely requires intubation
- **Prevention**
 - Avoid exposure to respiratory infections
 - Influenza vaccination
 - Hand hygiene
 - Breastfeeding supports immunity in infants

Bacterial Diseases of the Lower Respiratory System

Pneumonia

- Pneumonia is an infection that **inflames** the **air sacs** in one or both lungs.
 - Inflammation of the **Lung Parenchyma**
- Air sacs may fill with **pus** or **fluid**, causing **cough, fever, and difficulty breathing**.
- Can be caused by **bacteria, viruses, fungi, or aspiration**.
 - **Most cases** are caused by **bacteria** (adults).
- **Types of Pneumonia**
 - Community-acquired pneumonia (**CAP**)
 - Hospital-acquired pneumonia (**HAP**)
 - Ventilator-associated pneumonia (**VAP**)
 - VAP is a lung infection that develops in patients receiving **mechanical ventilation** (breathing machines) for at least 48 hours.
 - **Aspiration pneumonia**
 - Aspiration pneumonia is a lung infection that occurs when **fluids, food, or other foreign materials** enter the lungs.

Pneumonia...

- **Atypical pneumonia**, also known as “**walking pneumonia**” is a type of lung infection (pneumonia) caused by bacteria **different from the common ones** that typically cause the illness.
 - The term "atypical" refers to both the **causative organisms** and **the way the illness presents**.
 - The symptoms of atypical pneumonia tend to be **milder** and **develop more gradually** compared to typical bacterial pneumonia.
 - **Causative agents:** *Mycoplasma pneumoniae*, *Chlamydophila pneumoniae*, *Legionella pneumophila*

Comparison of Typical vs. Atypical Pneumonia

Feature	Typical Pneumonia	Atypical Pneumonia
Onset	Sudden	Gradual
Cough	Productive (purulent)	Dry/non-productive
Fever	High (>38.5°C)	Low-grade or absent
CXR Findings	Lobar consolidation. Unilateral.	Patchy diffuse (interstitial) infiltrates. Bilateral.
Extrapulmonary Signs	Rare	Common (e.g., rash, GI)
Etiology	<i>Streptococcus pneumoniae</i> , <i>Haemophilus influenzae</i> .	<i>Mycoplasma pneumoniae</i> , <i>Chlamydophila pneumoniae</i> , <i>Legionella</i> .

Pneumonia...

Common Causative Agents

- **Bacterial Pneumonia:** *Streptococcus pneumoniae* (most common), *Haemophilus influenzae*, *Klebsiella pneumoniae*
 - **Less common:** *Staphylococcus aureus*, *Pseudomonas aeruginosa*, *Escherichia coli*, *Acinetobacter baumannii*
 - May occur **primarily** or **secondary to viral infection** (e.g., flu or cold)
 - **Usually more severe** than viral pneumonia
- **Viral Pneumonia:** Influenza virus, RSV, SARS-CoV-2
 - Often **mild** and may resolve spontaneously
 - Can cause **serious illness** in infants, elderly, or immunocompromised
 - May **predispose** to **secondary bacterial pneumonia**
- **Fungal Pneumonia:** *Pneumocystis jirovecii* (in immunocompromised)

Pneumonia...

Clinical Manifestations

- Cough (productive or dry)
- Fever and chills
- Shortness of breath
- Chest pain
- Fatigue and weakness
- Crackles or rales on auscultation
 - **Abnormal breath sounds** that can be heard during pneumonia.

Pneumonia...

- **Diagnosis**
 - **Sputum Gram stain and culture**
 - Physical examination and history
 - Chest X-ray
- **Treatment**
 - **Antibiotics:** for bacterial pneumonia
 - **Antivirals:** if viral cause is identified
 - Supportive care: oxygen, fluids, antipyretics, rest
 - Hospitalization in severe cases
- **Complications**
 - Pleural effusion
 - Lung abscess
 - Respiratory failure
 - Sepsis

Pneumonia...

- **Prevention**
 - **Vaccination** (Pneumococcal, Influenza, COVID-19)
 - Smoking cessation
 - Hand hygiene
 - Proper care for chronically ill and elderly patients

Pertussis

- Pertussis (**Whooping Cough**) is a **highly contagious** respiratory disease caused by **Bordetella pertussis**.
 - A vaccine-preventable respiratory infection
 - B. pertussis is a **Gram-negative**, aerobic, encapsulated **coccobacillus**
- Characterized by severe **coughing fits** followed by a “**whooping**” sound.
- Affects people of all ages but is **most severe in infants**.
- **Transmission & Incubation**
 - Spread through **respiratory droplets** from coughing or sneezing.
 - Incubation period: **7–10 days (up to 21 days)**.
 - Most contagious during **early (catarrhal)** stage.

Pertussis...

- **Stages of Illness**
 1. **Catarrhal Stage** (1–2 weeks):
 - Mild cough, runny nose, low fever
 2. **Paroxysmal Stage** (2–6 weeks):
 - Severe, **violent coughing** fits
 - Inspiratory "whoop"
 - Post-tussive vomiting (vomiting induced by coughing)
 3. **Convalescent Stage** (weeks to months):
 - Gradual recovery

Pertussis...

- **Diagnosis**
 - Clinical history and symptoms
 - **Nasopharyngeal swab for PCR or culture (Bordet-Gengou medium)**
 - CBC: **Lymphocytosis** may be present
- **Treatment**
 - **Macrolide antibiotics:** azithromycin, erythromycin
 - **Supportive care:** hydration, oxygen if needed
 - **Early treatment limits** spread but may not alter course once paroxysms begin
- **Prevention**
 - **DPT vaccine** for children
 - Vaccination of pregnant women for newborn protection
 - Early case detection and isolation

Legionnaires' Disease

- Caused by *Legionella pneumophila*
- **Gram-negative**, aerobic **bacillus**
- First outbreak: **1976**, Philadelphia (American Legion Convention)
- **Transmission & Risk Factors**
 - **Inhalation of aerosolized water droplets** (cooling towers, showers)
 - No person-to-person transmission
 - Risk: elderly, smokers, immunocompromised
- **Pathogenesis**
 - Invades **alveolar macrophages**
 - Replicates within **phagosomes**
 - Causes **inflammation** and **lung damage**

Legionnaires' Disease...

- **Clinical Features**
 - Incubation: **2–10 days**
 - **Symptoms:** high fever, cough, dyspnea, myalgia, GI symptoms
 - May progress to **severe pneumonia**
- **Diagnosis**
 - **Urinary antigen** test
 - Culture on **Buffered Charcoal Yeast Extract (BCYE)** agar
 - PCR and serology
- **Treatment and Prevention**
 - **Antibiotics:** macrolides (azithromycin), fluoroquinolones
 - **Prevention:** disinfection of water systems
 - **No vaccine** available

Inhalational Anthrax

- Caused by *Bacillus anthracis*
- **Gram-positive, spore-forming rod**
- **Bioterrorism agent**; rare but deadly
- **Transmission & Risk Factors**
 - **Inhalation of spores** (from contaminated **animal products** or lab exposure)
 - **Not spread person-to-person**
 - **Risk**: workers handling animal hides, military, lab staff
- **Pathogenesis**
 - Spores taken up by **alveolar macrophages**
 - **Germinate** and spread to **lymph nodes**
 - **Toxins: Edema toxin, Lethal toxin** → systemic effects

Inhalational Anthrax...

- **Clinical Features**

- Incubation: **1–7 days**
- **Biphasic**: flu-like symptoms → **sudden respiratory distress**, sepsis
- **Widened mediastinum** (CXR hallmark)

- **Diagnosis**

- **Blood culture** (early)
- PCR and ELISA for toxin genes
- Chest imaging: mediastinal widening

- **Treatment and Prevention**

- **High-dose IV antibiotics**: ciprofloxacin + clindamycin or doxycycline
- **Antitoxin** (raxibacumab) in severe cases
- **Vaccine**: available for **high-risk** groups

Viral Diseases of the Lower Respiratory System

Influenza

Introduction

- **Acute respiratory viral infection** caused by **influenza viruses**.
- Causes **annual epidemics** and **occasional pandemics**.
- 3–5 million severe cases annually; 290,000–650,000 deaths.
- **Transmission:** Respiratory droplets, aerosols, contaminated surfaces (fomites).

Classification & Types

- **Family:** *Orthomyxoviridae*
- **Types:**
 - **Influenza A:** Causes **pandemics** (subtypes based on Hemagglutinin [H] & Neuraminidase [N])
 - **Influenza B:** Seasonal epidemics, less severe
 - **Influenza C:** Mild illness
- **Reservoirs:** Humans, birds, pigs (**influenza A is zoonotic**).

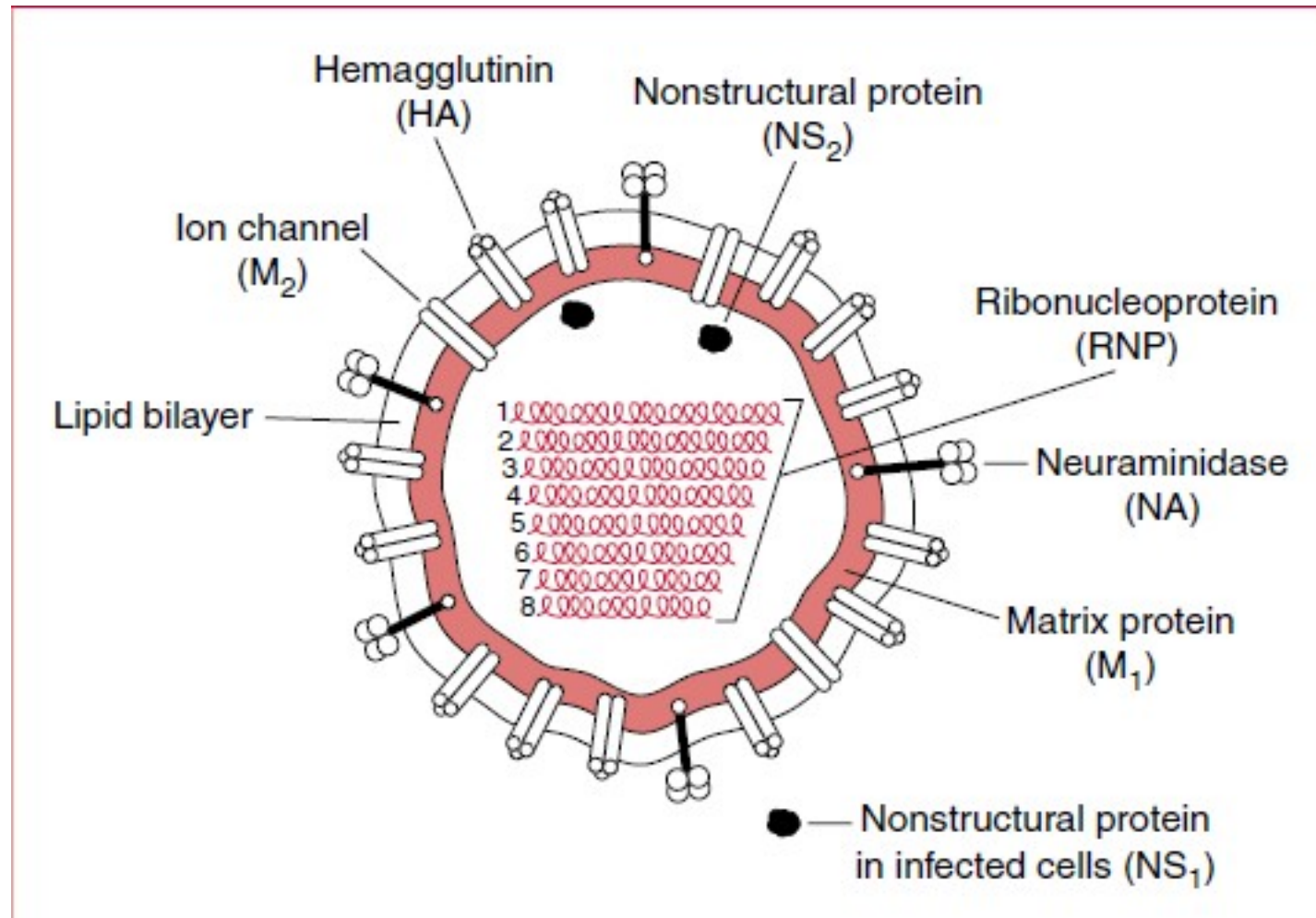
Influenza...

Viral Structure

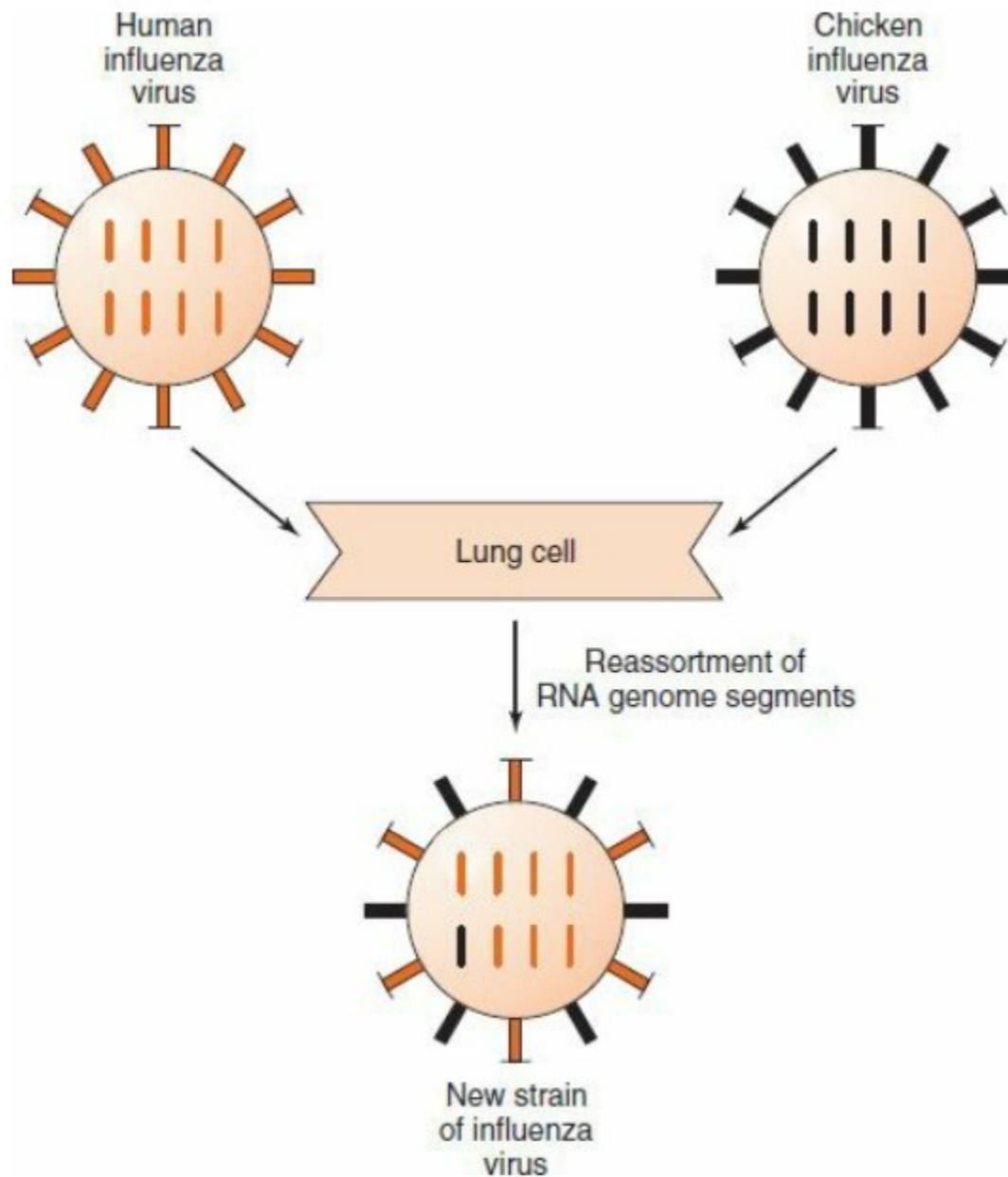
- **Enveloped**, negative-sense, **segmented** RNA virus (**8 segments**).
- **Surface glycoproteins:**
 - **Hemagglutinin (HA):** Attachment, entry
 - **Neuraminidase (NA):** Viral release from host cell
- **Other proteins:** M2 ion channel, NP, polymerase complex (PA, PB1, PB2).

Antigenic Variation

- **Antigenic drift:**
 - Minor mutations in HA/NA genes
 - Responsible for **annual epidemics**
- **Antigenic shift:**
 - **Reassortment** of gene segments between different strains
 - Responsible for **pandemics** (e.g., 1918 H1N1, 2009 H1N1).



Diagrammatic view of influenza A virus.



Antigenic shift in influenza virus. A human strain of influenza

Influenza...

Clinical Features

- **Incubation:** 1–4 days
- **Symptoms:**
 - Sudden onset high fever
 - Headache, myalgia, malaise (“flu-like”)
 - Dry cough, sore throat, rhinorrhea
- **Complications:**
 - Viral pneumonia
 - Secondary bacterial pneumonia
 - **Reye’s syndrome** (children, aspirin use)
 - Exacerbation of chronic diseases.

Influenza...

Diagnosis

- **Clinical suspicion** during outbreak season.
- **Rapid diagnostic tests:** Antigen detection (limited sensitivity).
- **RT-PCR: Gold standard**, highly sensitive and specific.
- **Viral culture:** Reference method, used in research/surveillance.

Treatment

- **Supportive therapy:** Rest, fluids, antipyretics.
- **Antivirals:**
 - **Neuraminidase inhibitors:** Oseltamivir, Zanamivir
 - **M2 inhibitors:** Amantadine, Rimantadine (limited use due to resistance).
- Best if started **within 48 hrs** of symptom onset.

Influenza...

Prevention

- **Annual vaccination** (inactivated, live-attenuated, recombinant).
- **Target groups:** elderly, children, healthcare workers, immunocompromised.
- **Non-pharmacological measures:** Hand hygiene, masks, isolation of cases.

Pandemic Influenza

- **History of major pandemics:**
 - 1918 “**Spanish flu**” (H1N1) → ~50 million deaths
 - 1957 “**Asian flu**” (H2N2)
 - 1968 “**Hong Kong flu**” (H3N2)
 - 2004 “**Bird Flu**” (H5N1)
 - 2009 “**Swine flu**” (H1N1)
- **Characteristics:** High attack rate, **antigenic shift**, global spread.

Acute Bronchitis

- Inflammation of the bronchial tubes
- Common in **all age groups**, especially **adults**
- **Self-limiting**, lasting < 3 weeks
- **Etiology**
 - **Mostly viral (90%)**
 - Influenza A & B, RSV, Rhinovirus, Coronavirus
 - **Bacterial causes (rare):** Mycoplasma pneumoniae, Chlamydia pneumoniae, Bordetella pertussis
- **Clinical Features**
 - **Cough** (often productive), chest discomfort
 - Wheezing, rhonchi (abnormal lung sounds)
 - Mild fever
 - No signs of consolidation (vs. pneumonia)

Bronchiolitis

- Inflammation of the **small airways (bronchioles)**
- Primarily affects **infants and young children (<2 years)**
- **Seasonal outbreaks**, often in winter
- **Etiology**
 - **Most common cause:** **Respiratory Syncytial Virus (RSV)**
 - **Others:** Parainfluenza, Adenovirus, Rhinovirus, Human metapneumovirus (HMPV)
- **Clinical Features**
 - Initial upper respiratory symptoms → **cough, wheezing, dyspnea**
 - Low-grade fever, nasal flaring (nostril widening), intercostal retractions
 - **Apnea** may occur in infants
 - Risk of dehydration due to feeding difficulty

Hantavirus Pulmonary Syndrome (HPS)

- Caused by **Hantavirus**, -ssRNA virus, family Bunyaviridae,
- First recognized in the **U.S. in 1993**
- Associated with **rodent exposure** (e.g. mice, rats)
- **Transmission & Risk Factors**
 - **Inhalation of aerosolized rodent excreta**
 - **No person-to-person** transmission
 - **Risk:** rural areas, rodent-infested homes
- **Pathogenesis**
 - Virus infects **endothelial cells**
 - Increased **vascular permeability** → **pulmonary edema**
 - Can lead to **cardiopulmonary failure**

Hantavirus Pulmonary Syndrome...

- **Clinical Features**
 - Incubation: **1–5 weeks**
 - **Prodrome**: fever, myalgia, nausea
 - **Cardiopulmonary phase**: cough, dyspnea, hypotension
- **Diagnosis & Management**
 - **Serology** (IgM, IgG), RT-PCR
 - **Chest X-ray**: bilateral infiltrates, interstitial edema
 - **Supportive care**: ICU, mechanical ventilation
 - **Mortality rate**: **~35–40%**

Fungal Diseases of the Respiratory System

Coccidioidomycosis

- Coccidioidomycosis is also known as **Valley fever**.
- **Causative agent:** *Coccidioides immitis* or *C. posadasii*
 - **Systemic endemic** pathogen
- **Type: Dimorphic fungus**
 - Mold (arthroconidia) in soil
 - Spherules in tissue
- **Epidemiology:** Southwestern USA, Mexico
- **Transmission:** Inhalation of **arthroconidia** (spore) from soil
- **Clinical Features:**
 - **Flu-like illness**, cough, chest pain
 - May progress to chronic or disseminated disease
- **Diagnosis:** Spherules in tissue, serology
- **Treatment:** Fluconazole or amphotericin B

Blastomycosis

- **Causative agent:** *Blastomyces dermatitidis*
 - **Systemic endemic** pathogen
- **Type:** **Dimorphic fungus**
 - Mold in environment
 - Yeast (broad-based budding) in tissue
- **Epidemiology:** Mississippi & Ohio River Valleys
- **Transmission:** **Inhalation of conidia**
- **Clinical Features:**
 - **Pneumonia-like** symptoms
 - May disseminate to skin, bones
- **Diagnosis:** Broad-based **budding yeast** in **sputum** or **biopsy**
- **Treatment:** Itraconazole or amphotericin B

Histoplasmosis

- Causative agent: *Histoplasma capsulatum*
 - Systemic endemic pathogen
- Type: **Dimorphic fungus**
 - Mold in soil
 - Yeast (intracellular) in host macrophages
- **Epidemiology: Soil with bird/bat droppings** (caves, chicken coops)
- **Transmission: Inhalation of microconidia**
- **Clinical Features:**
 - Asymptomatic or **mild flu-like** illness
 - **Severe cases in immunocompromised:** disseminated histoplasmosis
- **Diagnosis:** Intracellular **yeast in macrophages**, **antigen** detection
- **Treatment:** Itraconazole or amphotericin B

Pneumocystis Pneumonia (PCP)

- Causative agent: *Pneumocystis jirovecii*
 - Opportunistic pathogen
 - Atypical fungus (not dimorphic, not culturable)
- Epidemiology: Worldwide; mostly in immunocompromised (e.g. AIDS)
- Transmission: Airborne (not fully understood)
- Clinical Features:
 - Progressive **dyspnea**, nonproductive cough, fever
 - Hypoxemia, bilateral interstitial infiltrates on X-ray
- Diagnosis: Silver stain or PCR from **sputum/BAL**
- Treatment: TMP-SMX (trimethoprim-sulfamethoxazole)
- Prophylaxis: For CD4 < 200 cells/mm³ in HIV patients

Selected Respiratory Microbe

1. *Streptococcus pyogenes*
2. *Streptococcus pneumoniae*
3. *Corynebacterium diphtheriae*
4. *Haemophilus influenzae*
5. Parainfluenza viruses
6. Respiratory syncytial virus (RSV)

Streptococcus pyogenes

Introduction

- *Streptococcus pyogenes* = **Group A β -hemolytic Streptococcus (GAS)**
- **Gram-positive cocci** arranged in chains
- **Facultative anaerobe**
- Important cause of **upper respiratory tract infections**
 - Common cause of **bacterial pharyngitis**

Morphology & Characteristics

- Gram-positive, non-motile, non-spore-forming cocci
- **β -hemolytic** on blood agar
- **Catalase-negative**
- **Bacitracin-sensitive**
- **Lancefield Group A antigen**



Streptococcus pyogenes...

Virulence Factors

- **M protein**: antiphagocytic, major virulence factor
- **Streptolysins (O, S)**: cause β -hemolysis, cytotoxic
- **Streptokinase**: dissolves clots → spreads infection
- **Hyaluronidase**: degrades connective tissue
- **Pyrogenic exotoxins**: cause rash and toxic effects (e.g., scarlet fever)

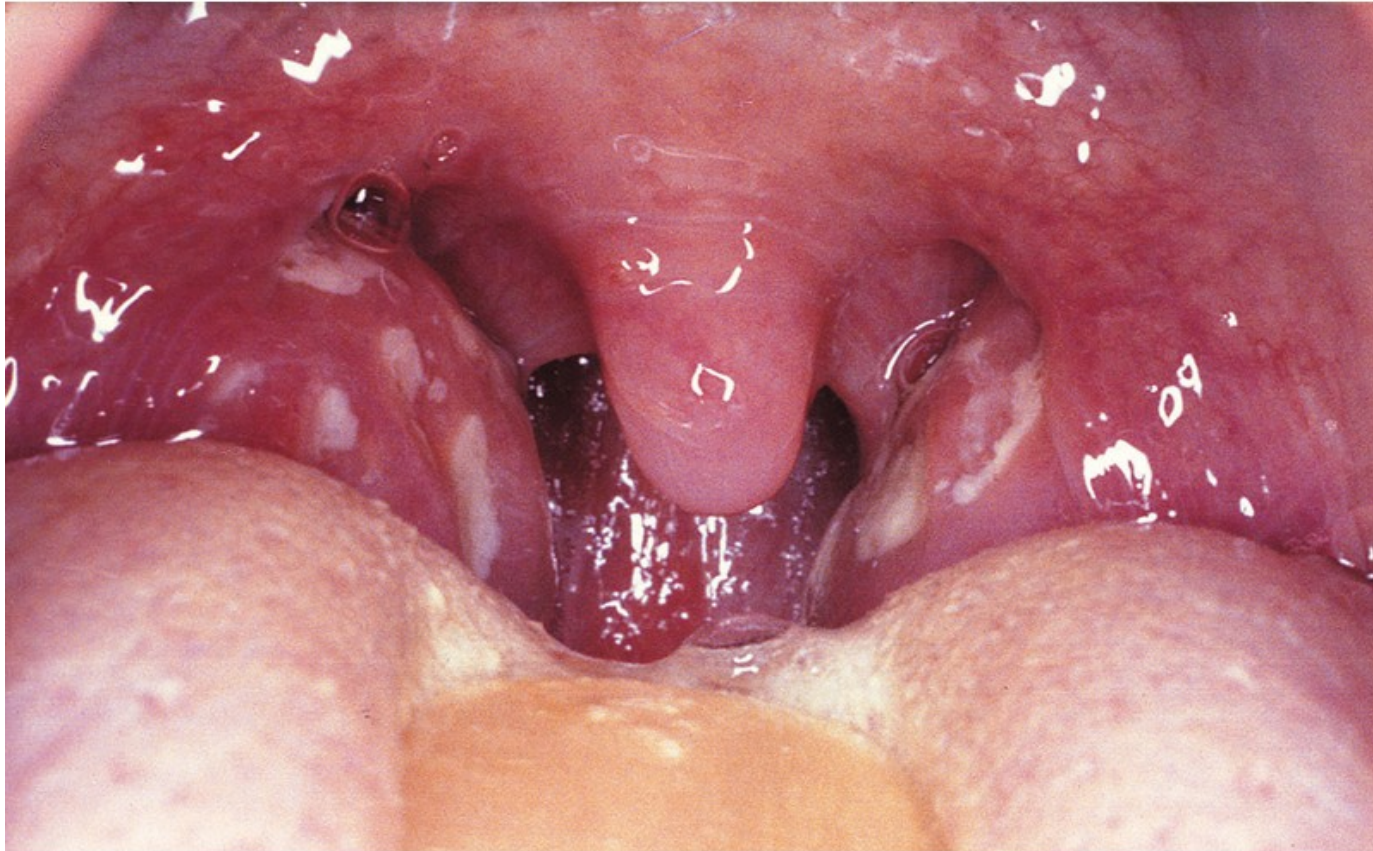
Epidemiology

- Reservoir: **Human throat and skin**
- Transmission: **Respiratory droplets** and **direct contact**
- Common in **school-aged children**
- Occurs more in **crowded settings**

Streptococcus pyogenes...

Clinical Manifestations

- **Streptococcal pharyngitis (strep throat):**
 - Sore throat, fever, tonsillar **exudate**, tender cervical lymph nodes
- **Scarlet fever:**
 - Erythematous rash, strawberry tongue (due to **exotoxin**)
- **Complications:**
 - *Suppurative*: otitis media, sinusitis, abscess
 - *Non-suppurative*: **rheumatic fever**, post-streptococcal glomerulonephritis



© From Farrar W.E., Woods M.J., Innes J.A.: Infectious Diseases; Text and Color Atlas, ed 2. London, Mosby Europe, 1992.

Fig. Pharyngitis

Streptococcus pyogenes...

Laboratory Diagnosis

- **Specimen:** Throat swab
- **Tests:**
 - Culture on **blood agar** → β -hemolysis
 - **Bacitracin sensitivity** (zone of inhibition)
 - **Rapid antigen detection test (RADT)**
 - **ASO titer** for recent infection (rheumatic fever)

Treatment and Prevention

- **Drug of choice:** Penicillin or Amoxicillin
- For allergy: Macrolides (e.g., erythromycin)
- Supportive therapy for symptoms
- **Prevention:**
 - Early treatment of pharyngitis
 - Avoid contact with infected persons
 - No effective vaccine yet

Streptococcus pneumoniae

Introduction

- *Streptococcus pneumoniae* (Pneumococcus) is a **Gram-positive, encapsulated diplococcus**.
- Major cause of **community-acquired pneumonia**
 - Also causes **otitis media, sinusitis, meningitis, and bacteremia**
 - *S. pneumoniae* is a **leading cause of bacterial pneumonia and meningitis**.
- Exists as part of the **normal nasopharyngeal flora** in healthy individuals.

Morphology and Characteristics

- **Gram-positive, lancet-shaped diplococci**
- **Encapsulated** – capsule is the **main virulence factor**
- **α -hemolytic** on blood agar (greenish zone)
- **Catalase-negative**
- **Optochin-sensitive** and **bile soluble**

Streptococcus pneumoniae...

Virulence Factors

- **Polysaccharide capsule:** antiphagocytic, prevents opsonization
- **Pneumolysin:** damages epithelial cells and cilia
- **Autolysin:** contributes to inflammation and release of virulence factors
- **IgA protease:** facilitates colonization of mucosal surfaces

Epidemiology

- Reservoir: **Human nasopharynx**
- Transmission: **Respiratory droplets**
- Common in **infants, elderly, and immunocompromised individuals**
- Infections peak during **cold and dry seasons**

Streptococcus pneumoniae...

Clinical Manifestations

- **Pneumonia:** sudden fever, chills, productive cough with rusty sputum, chest pain
- **Otitis media:** ear pain, fever, irritability (common in children)
- **Sinusitis:** facial pain, nasal discharge
- **Meningitis:** headache, neck stiffness, photophobia
- **Bacteremia/sepsis:** in severe cases

Streptococcus pneumoniae...

Laboratory Diagnosis

- **Specimens:** Sputum, blood, CSF, or middle ear aspirate
- **Microscopy:** Gram-positive diplococci in pairs
- **Culture:** α -hemolytic colonies on blood agar
- **Optochin test:** sensitive
- **Bile solubility test:** positive
- **Quellung reaction:** capsular swelling with specific antisera
- **Antigen detection:** in CSF or urine (latex agglutination)

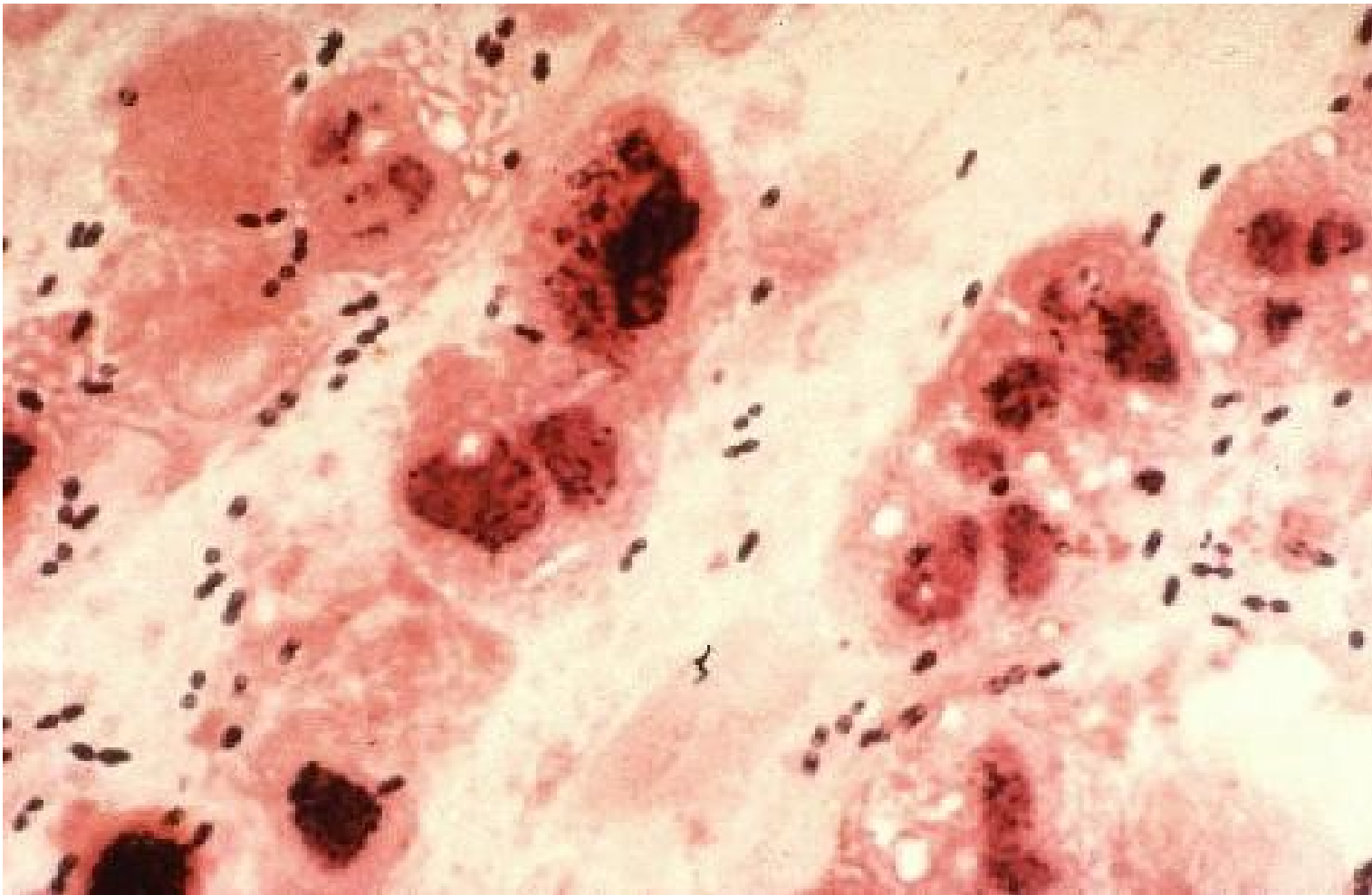


Figure: Gram Stain of a film of **sputum** from a case of **lobar pneumonia**

Most bacteria (e.g., *Streptococcus pneumoniae*, *Klebsiella pneumoniae*, *Haemophilus influenzae*) that cause lung infections (e.g., pneumonia) produce a **capsule** that can **prevent phagocytosis by the alveolar macrophage**.

Streptococcus pneumoniae...

Treatment and Prevention

- **Antibiotics:** Penicillin (if sensitive), or third-generation cephalosporins
- Alternatives: Macrolides or fluoroquinolones (if resistant)
- **Vaccines:**
 - **PCV13** (conjugate vaccine) – for **children** and high-risk adults
 - **PPSV23** (polysaccharide vaccine) – for **adults** and elderly
- **Preventive measures:** Respiratory hygiene, vaccination of at-risk groups

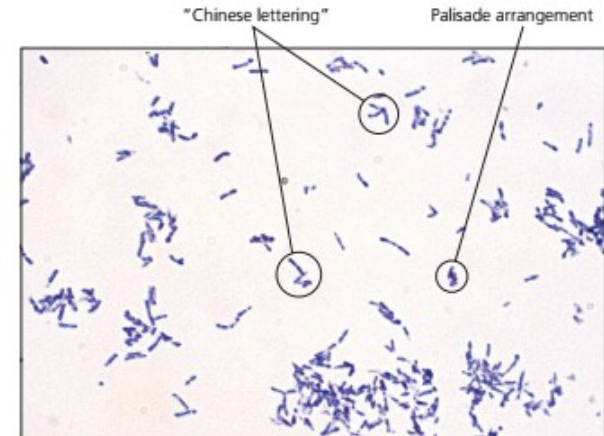
Corynebacterium diphtheriae

Introduction

- *Corynebacterium diphtheriae* → **causative agent of diphtheria**
- **Gram-positive, non-spore-forming rod**
- **Produces potent exotoxin** → causes local and systemic disease
- Disease primarily affects **throat and upper respiratory tract**

Morphology & Characteristics

- **Gram-positive, club-shaped bacilli**
- Arranged in **palisades or Chinese-letter forms**
- **Non-motile, non-capsulated, non-spore-forming**
- **Aerobic / facultative anaerobe**
- **Metachromatic (Babes–Ernst) granules** seen with special stains (e.g., Albert's stain)



Corynebacterium diphtheriae...

Virulence Factors

- **Diphtheria toxin** (main virulence factor):
 - Encoded by **tox gene** (carried by **bacteriophage**)
 - **A-B toxin**:
 - **A subunit** → inhibits protein synthesis (ADP-ribosylates EF-2)
 - **B subunit** → binds to host cell receptors
- **Surface adhesins** help colonization of throat epithelium

Epidemiology

- **Reservoir**: Humans (asymptomatic carriers)
- **Transmission**: Respiratory droplets, direct contact
- **Common in unvaccinated individuals**
- **Incubation period**: 2–6 days

Corynebacterium diphtheriae...

Clinical Manifestations

- **Respiratory diphtheria:**
 - Sore throat, fever, difficulty swallowing
 - **Pseudomembrane** on tonsils, pharynx, or larynx
 - “**Bull neck**” due to lymphadenopathy and edema
- **Complications:**
 - Myocarditis
 - Peripheral neuropathy
 - Airway obstruction



This child has diphtheria resulting in a **thick gray coating over back of throat**. This coating can eventually expand **down through airway** and, if not treated, the child could die from **suffocation**.

Corynebacterium diphtheriae...

Laboratory Diagnosis

- **Specimen:** Throat or nasal swab from membrane edge
- **Microscopy:** Gram-positive bacilli, metachromatic granules
- **Culture:**
 - *Loeffler's medium* or *Tinsdale (tellurite) agar* → black colonies
- **Toxin detection:**
 - *Elek test* (in vitro toxigenicity test)
 - PCR for *tox* gene
 - ELISA for toxin detection

Corynebacterium diphtheriae...

Treatment and Prevention

- **Treatment:**
 - *Diphtheria antitoxin* (neutralizes circulating toxin)
 - *Antibiotics*: Penicillin or Erythromycin
 - Supportive care for airway and cardiac complications
- **Prevention:**
 - *DTP/DTaP* vaccine (contains diphtheria toxoid)
 - Booster every 10 years
 - Isolation of cases and prophylaxis of contacts.

Haemophilus influenzae

Introduction

- *Haemophilus influenzae* → small **Gram-negative coccobacillus**
- Causes **upper and lower respiratory tract infections**
- Two forms:
 - **Encapsulated (type b – Hib)**
 - **Non-encapsulated (nontypeable strains)**
- Important pathogen in **children and immunocompromised individuals**

Morphology & Characteristics

- Small, pleomorphic **Gram-negative coccobacilli**
- **Non-motile, non-spore-forming**
- **Requires X (hemin) and V (NAD) factors** for growth
- Grows on **chocolate agar**, not on blood agar
- Facultative anaerobe

Haemophilus influenzae...

Virulence Factors

- **Polyribosylribitol phosphate (PRP) capsule** (major virulence factor) – antiphagocytic (Type b)
- **IgA protease** – destroys mucosal IgA
- **Endotoxin (LPS)** – inflammation and tissue damage
- **Adhesins** – help attachment to respiratory epithelium

Epidemiology

- **Reservoir:** Human nasopharynx
- **Transmission:** Respiratory droplets, close contact
- **Hib** mainly affects **infants and children <5 years**
- Incidence greatly reduced by **Hib vaccination**

Haemophilus influenzae...

Clinical Manifestations

- **Respiratory infections: (nontypeable strains)**
 - Otitis media, sinusitis, bronchitis, **pneumonia**
- **Hib (invasive) infections:**
 - **Epiglottitis** → sudden airway obstruction, drooling, stridor
 - Meningitis
 - Septicemia
- **Others:** Conjunctivitis, cellulitis

Haemophilus influenzae...

Laboratory Diagnosis

- **Specimens:** Sputum, blood, CSF, or nasopharyngeal swab
- **Microscopy:** Small Gram-negative coccobacilli
- **Culture:** Chocolate agar with X and V factors
- **Satellite phenomenon:** growth around *S. aureus* on blood agar
- **Antigen detection:** PRP capsule antigen in CSF or urine (Hib)

Treatment and Prevention

- **Treatment:**
 - *Ceftriaxone* or *ampicillin* (if sensitive)
 - Supportive therapy for airway obstruction
- **Prevention:**
 - **Hib conjugate vaccine** (part of routine childhood immunization)
 - Rifampicin prophylaxis for close contacts
 - Avoid smoking (reduces mucosal damage)

Parainfluenza Viruses

Introduction

- Parainfluenza viruses are **enveloped, single-stranded, negative-sense RNA viruses**.
- Belong to the **Paramyxoviridae** family.
- Cause **upper and lower respiratory tract infections**, especially in **children**.

Classification

- **Human Parainfluenza Virus (HPIV) types:**
 - **HPIV-1** → **Major cause of croup** (*laryngotracheobronchitis*)
 - **HPIV-2** → Also causes *croup* and **upper respiratory infections**
 - **HPIV-3** → Causes **bronchiolitis** and **pneumonia** in infants
 - **HPIV-4** → Milder infections, less common

Parainfluenza Viruses...

Epidemiology

- **Worldwide distribution**
- **Common in children under 5 years**
- **Transmission: Respiratory droplets and close contact**
- **Peak incidence: Fall and early winter**

Clinical Manifestations

- **Mild URTI:** cough, coryza, sore throat
- **Croup:** barking cough, hoarseness, stridor
- **Bronchiolitis / Pneumonia:** fever, wheezing, respiratory distress
- **In adults:** mild cold-like illness

Parainfluenza Viruses...

Diagnosis

- **Specimen:** Nasopharyngeal swab or aspirate
- **Laboratory methods:**
 - RT-PCR (most sensitive)
 - Direct antigen detection (immunofluorescence)
 - Virus isolation in cell culture
 - Serology (paired sera – rise in antibodies)

Treatment and Prevention

- **No specific antiviral therapy**
- **Supportive care:** humidified air, hydration, oxygen if needed
- **Corticosteroids** for severe croup
- **Prevention:**
 - Hand hygiene
 - Avoid close contact with infected individuals
 - No licensed vaccine yet

Respiratory Syncytial Virus (RSV)

- **-ssRNA** virus, enveloped, family Paramyxoviridae
- **Major cause** of **bronchiolitis** and **pneumonia** in infants
 - Also associated with: **laryngotracheobronchitis** and **common cold**
- Common in **winter** months
- **Transmission & Epidemiology**
 - **Droplet** and **contact** spread
 - **Nearly all children** infected by **age 2**
 - **Severe in infants**, elderly, and immunocompromised
- **Pathogenesis**
 - Infects **ciliated epithelial cells**
 - **Syncytium** formation (cell fusion)
 - Causes **airway obstruction** and inflammation

Respiratory Syncytial Virus Infection...

- **Clinical Features**
 - **In infants:** cough, wheezing, difficulty breathing
 - **In older children/adults:** mild cold-like symptoms
 - Can cause **apnea** in neonates (Temporary **cessation of breathing**).
- **Diagnosis & Management**
 - **Nasopharyngeal swab** for antigen/PCR
 - **Supportive care:** oxygen, hydration
 - Palivizumab (**monoclonal antibody**) for prevention in **high-risk infants**
 - **No vaccine** available yet

Tuberculosis (TB)

Introduction

- TB is a contagious bacterial infection caused by *Mycobacterium tuberculosis*.
 - Belongs to the **Mycobacterium tuberculosis complex (MTBC)**
 - **Other members:** *M. bovis*, *M. africanum*, *M. microti*
 - **Acid-fast bacillus (AFB)**
 - Discovered by **Robert Koch (1882)**
- Remains one of the **top 10 causes of death worldwide**

General Characteristics

- Obligate **aerobe**
- Non-motile, non-spore-forming
- **Slow-growing** (generation time: 15–20 hours)
- Acid-fast due to **mycolic acids** in cell wall (**Lipid-rich** cell wall)
- Resistant to **desiccation** and many **disinfectants**

Tuberculosis...

Epidemiology

- **Global burden:** ~10 million new cases/year
- **High prevalence in developing countries**
- **Transmission:** airborne droplets (coughing, sneezing, talking)
- **Risk factors:**
 - HIV/AIDS
 - Malnutrition
 - Overcrowding
 - Diabetes & immunosuppression

Tuberculosis...

Pathogenesis

- Entry via **inhalation of droplet nuclei** (< 5 μm in diameter)
- Bacilli reach **alveoli** → **engulfed by macrophages**
- Survive inside macrophages by **inhibiting phagosome-lysosome fusion**
- Formation of **granulomas** (tubercles)
- Outcomes:
 - **Latent TB infection (LTBI)** (→ **Secondary TB (Reactivation TB)**)
 - **Active TB disease** (**Primary TB, 5-10%**)
- **Primary TB** has generally been defined as onset of clinical disease **within 12 months of initial infection** with MTB.
 - **Secondary TB** usually occurs because of **reactivation** of **latent TB** infection or **reinfection**.

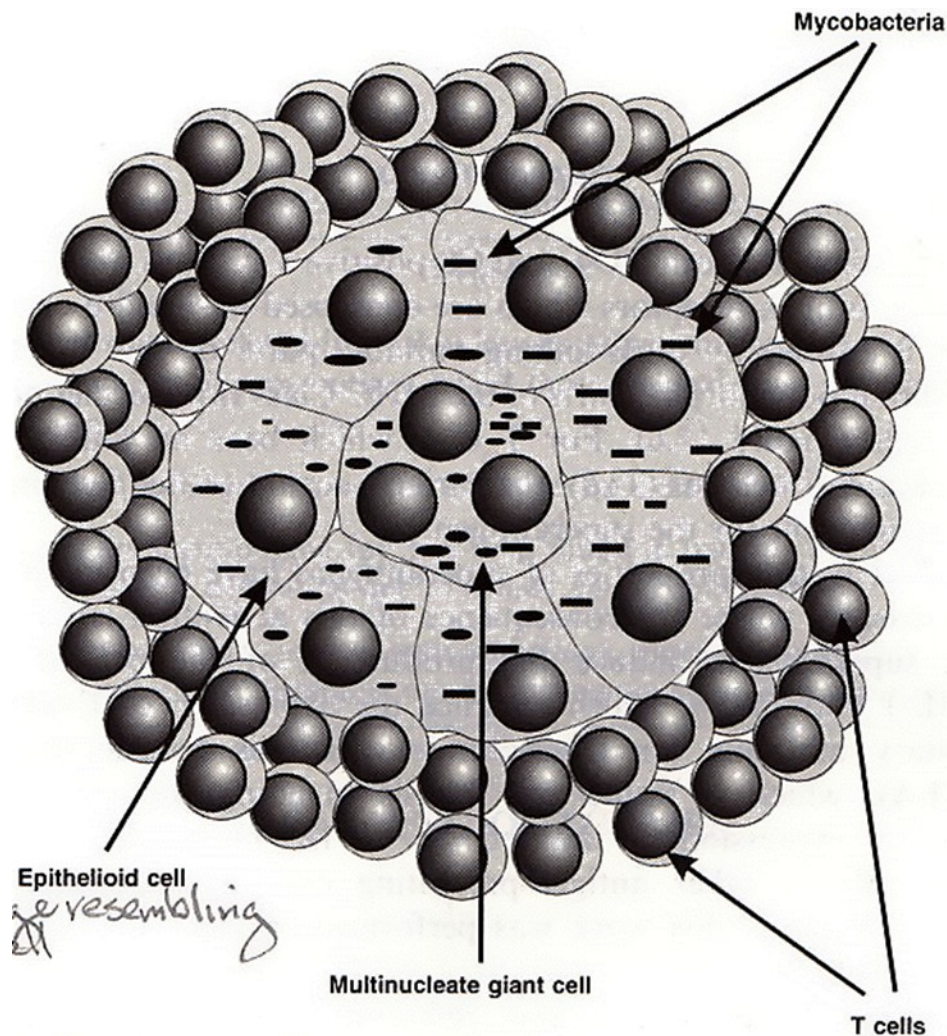


Diagram of a **Granuloma**

- **Epithelioid cells** are derivatives of **activated macrophages** resembling epithelial cells.
- Note: Ultimately a fibrin layer develops around granuloma (fibrosis), further “**walling off**” the lesion.
- Typical progression in pulmonary TB involves caseation, calcification and cavity formation.

Tuberculosis...

Clinical Features:

Pulmonary TB (PTB) (most common):

- Persistent **cough (>2 weeks)**
- Hemoptysis (blood in sputum)
- Night sweats, fever, weight loss (“consumption”)

Extrapulmonary TB (EPTB):

- **Lymph nodes**, pleura, bones/joints, genitourinary tract, CNS (meningitis), miliary TB (disseminated TB)

Tuberculosis...

Laboratory Diagnosis

1. Microscopy

- **Ziehl–Neelsen** stain (red bacilli on blue background)
- Auramine–rhodamine (fluorescent staining)

2. Culture (gold standard)

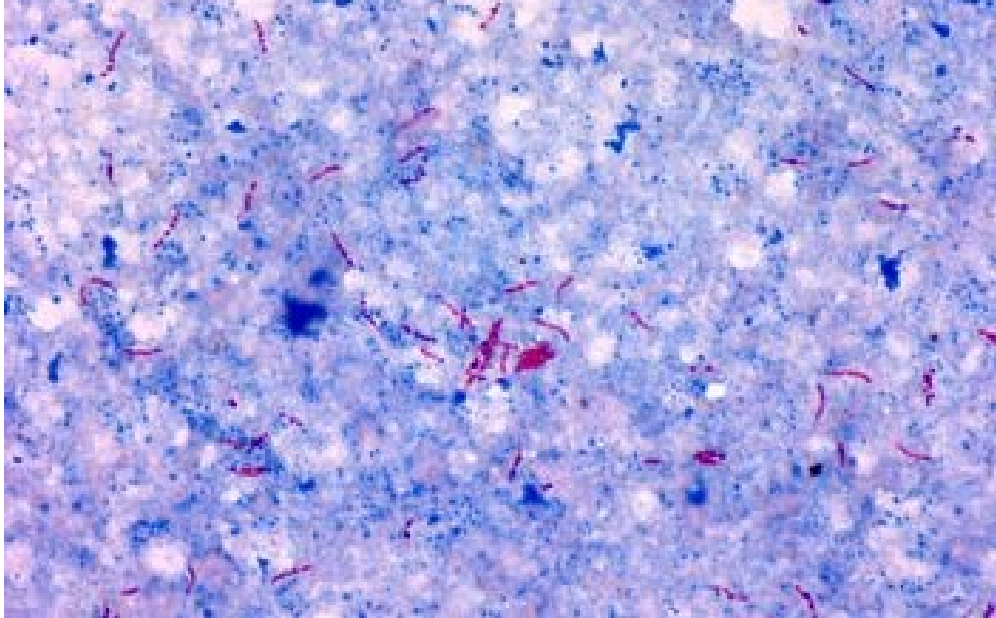
- **Löwenstein–Jensen (LJ)** medium (slow growth: 3–8 weeks)

3. Molecular tests

- **GeneXpert MTB/RIF** (rapid, detects rifampicin resistance)
- PCR

4. Immunological tests

- **Tuberculin skin test (Mantoux)**
- Interferon-gamma release assays (IGRAs)

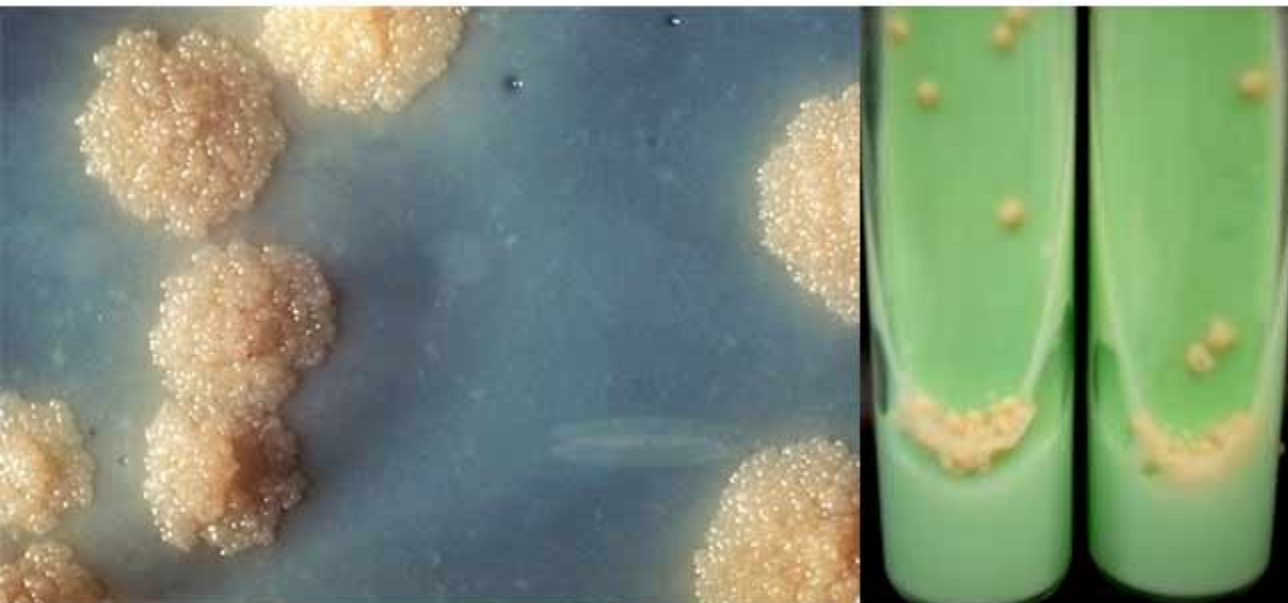


Red, straight or slightly curved rods, occurring singly or in small groups, may appear beaded

AURAMINE-RHODAMINE STAIN



AFB: **reddish-yellow** fluorescence
Background: black



Mtb produces **raised**, dry, **cream** (buff) colored colonies

Fig: Cultural Characteristics of *Mycobacterium tuberculosis*



Figure. The **GeneXpert MTB/RIF** assay

Tuberculosis...

Treatment

- Treatment requires **multiple drugs & prolonged therapy**
- **First-line anti-TB drugs (RIPE):**
 - Rifampicin
 - Isoniazid
 - Pyrazinamide
 - Ethambutol
- **Standard regimen (WHO):**
 - **Intensive phase** (2 months): RIPE
 - **Continuation phase** (4 months): RI
- Directly Observed Treatment, Short-course (**DOTS**) recommended

Tuberculosis...

Drug Resistance

- **MDR-TB:** Resistant to at least **INH & Rifampicin**
 - Requires **second-line drugs** (fluoroquinolones, injectables)
- **XDR-TB:** MDR + resistant to fluoroquinolones & ≥ 1 second-line injectable drug
 - **MDR/XDR-TB** = major global threat
- **Causes:** Poor adherence, inadequate regimens, interrupted supply.

Tuberculosis...

Prevention & Control

- **Early case detection & treatment**
- **BCG vaccine** (live attenuated *M. bovis*): Effective against **severe childhood TB**
- Respiratory hygiene, ventilation, masks
- Directly Observed Treatment, Short-course (**DOTS**) strategy

SARS

- SARS (Severe Acute Respiratory Syndrome) is a **viral respiratory illness** caused by the **SARS coronavirus (SARS-CoV)**.
- First identified in Asia in **February 2003**.
 - SARS was first discovered in **November 2002 in Guangdong Province, China**.
 - The outbreak was officially reported to the WHO in **February 2003**, and it quickly spread to several countries, becoming a global health concern.
- **Outbreaks:** Rapid spread in over **26 countries** with significant morbidity and mortality.
 - The outbreak was largely contained by **July 2003**, with over **8,000 cases** and **774 deaths** reported worldwide.

SARS...

Etiology

- **Causative Agent: SARS-CoV**, a +ssRNA helical virus, enveloped.
 - **Coronaviridae family**: SARS-CoV, MERS-CoV (2012), SARS-CoV-2.
- **Reservoir**: Thought to be **bats**; intermediary hosts include **civet cats**.

Transmission

- Respiratory droplets
- Close person-to-person contact
- Fomites

Pathogenesis

- Binds to **ACE2 receptors** in respiratory epithelium
- Causes **diffuse alveolar damage**
- **Cytokine storm** may contribute to severity

SARS...

Clinical Features

- **Incubation Period:** 2–7 days.
- **Symptoms:**
 - High fever ($>38^{\circ}\text{C}$)
 - Dry cough, difficulty breathing
 - Shortness of breath (**dyspnea**) → **hypoxia** (low levels of O_2 in the body's tissue).
 - Headache, muscle pain, fatigue
 - **Severe cases: Pneumonia and respiratory failure**
- **Complications:** Acute respiratory distress syndrome (**ARDS**), multi-organ failure.

SARS...

Diagnosis

- **Laboratory Tests:**
 - **RT-PCR** for SARS-CoV RNA (gold standard)
 - **Serological testing** (antibodies)
- **Imaging:** Chest X-ray or CT scan showing **atypical pneumonia**.
- **Clinical Criteria:** Travel history to affected regions or close contact with suspected cases.

Treatment

- Supportive Care:
 - **Oxygen therapy**
 - **Mechanical ventilation** in severe cases
- **Antiviral Drugs:** Limited efficacy; ongoing research.
- **Corticosteroids:** Used to reduce inflammation, though controversial.
- **Experimental Therapies:** Interferons, monoclonal antibodies.

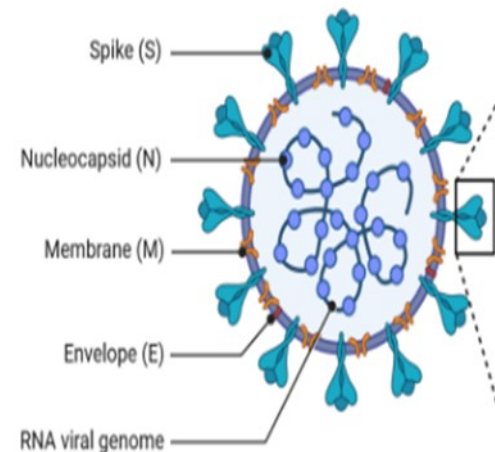
SARS...

Prevention

- Hygiene Measures:
 - **Frequent handwashing**
 - **Avoid touching the face**
- **Isolation:** Quarantining suspected and confirmed cases.
- **Protective Equipment:** Masks, gloves, and gowns for healthcare workers.
- **Travel Restrictions:** Limiting movement in affected areas during outbreaks.
- Human-to-human transmission of **SARS-CoV-1** is considered to have been **eradicated since 2004**, with no cases reported worldwide since then.

COVID-19

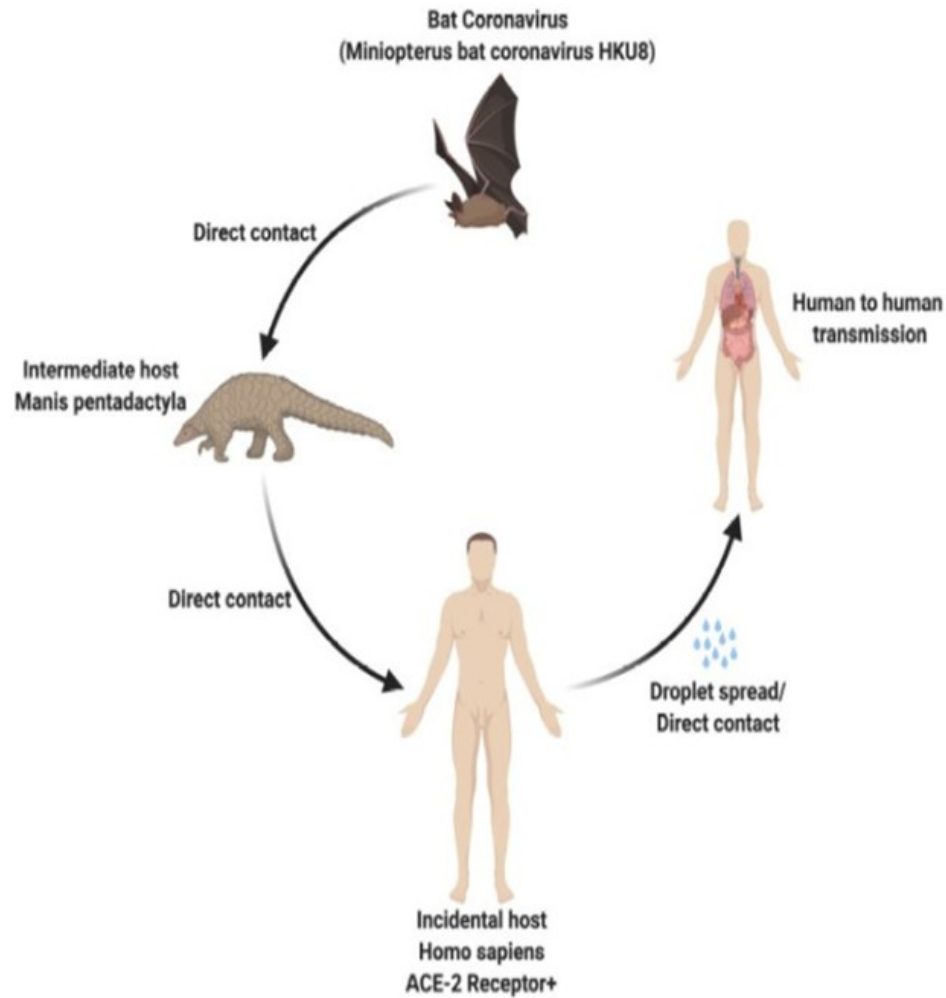
- COVID-19 (Coronavirus disease 2019) Caused by the **SARS-CoV-2** virus.
 - Family: **Coronaviridae** (enveloped, +ssRNA viruses).
- First identified in Wuhan, China, in **December 2019**.
 - Declared a **pandemic** by WHO in March 2020.
- Affected **millions** worldwide, causing significant health, economic, and social disruptions.
- The global number of COVID-19 cases and deaths (2019-2024):
 - Total Cases: **Over 700 million** confirmed cases worldwide.
 - Total Deaths: **Over 6.9 million** reported deaths globally.



Transmission

- Human-to-human transmission
 - Respiratory **droplets** from coughs, sneezes, or talking.
 - **Contact** with contaminated surfaces
- High transmission in **crowded** and **poorly ventilated** spaces

Transmission Cycle of SARS CoV 2



*Chinese **pangolin** (Manis pentadactyla)

COVID-19...

Symptoms

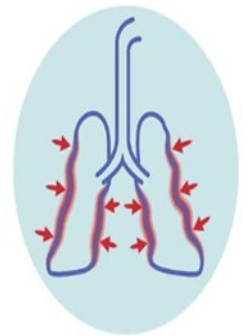
- Incubation period: **1-14 days (average 5 days)**.
- The virus is **most contagious** for the **first three days** after the appearance of symptoms.
- Common symptoms:
 - Fever, **cough**, fatigue
 - **Loss of taste or smell**
- Severe symptoms:
 - **Difficulty breathing, chest pain**
 - **Dyspnea** (Shortness of breath)
 - Confusion or loss of consciousness



Fever



Cough



Shortness of breath

COVID-19...

Risk Factors

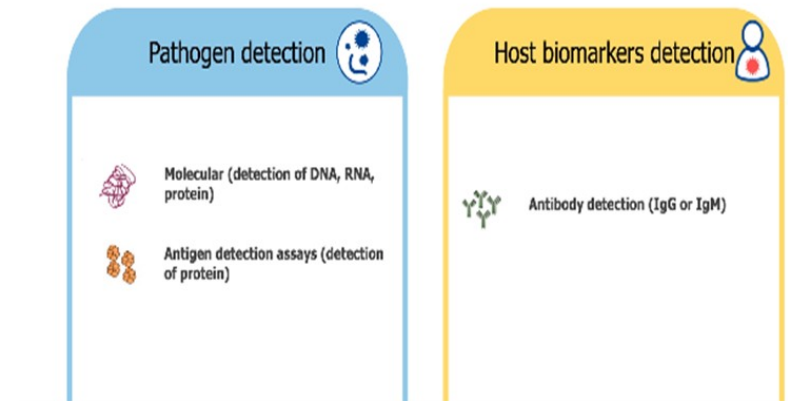
- The primary risk factor associated with coronavirus is **age** (**elderly** are at higher risk).
 - A case fatality rate of **0.32%** is seen in those aged **below 60 years**
 - **6.4%** in those aged **≥60 years**
 - As high as **13.4%** in those aged more than **80 years**.
- **Underlying disorders**
 - Hypertension, respiratory conditions (**chronic obstructive pulmonary disease**), diabetes, cardiovascular disease,
 - are found to be at more risk of developing severe symptoms.
- **Immunocompromised** individuals

COVID-19...

Diagnosis

- Common diagnostic tests:
 - **RT-PCR (gold standard)**
 - Rapid antigen tests
- **Imaging:**
 - Chest X-rays or CT scans in severe cases

Basic diagnostic methods



Treatment

- **Mild cases:** Symptomatic treatment (rest, hydration, fever management)
- **Severe cases:**
 - Hospitalization, **oxygen therapy**
 - Antiviral drugs (e.g., remdesivir)
 - Steroids for inflammation

COVID-19...

Prevention Measures

- **Key Strategies:**
 - Vaccination
 - Wear masks in public places
 - Maintain social distancing
 - Regular handwashing with soap and use of hand sanitizers
 - Avoid large gatherings

Vaccination

- **Available vaccines:** Pfizer-BioNTech, Moderna, AstraZeneca, Johnson & Johnson, etc.
- Proven **effective in reducing severe illness** and death.
- **Booster doses** for prolonged immunity

Challenges

- Vaccine hesitancy.
- Emergence of variants (e.g., **Delta, Omicron**).